GOVERNMENT OF TELANGANA
ELECTRICAL INSPECTORATE
(An IS/ISO 9001: 2015 Certified Department)

BUREAU OF INDIAN STANDARDS — IS / ISO 9001 — QMS

From

The Chief Electrical Inspector to Government
5th Floor
R&B Buildings
A.C. Guards
Masabtank
Hyderabad-500004.
Phone No: **040-23453023**

To

M/s: **SRI RAJ IRON & STEEL PRODUCTS PVT LTD**
Street: CHANDAMPET
H.No/D.No.: SY NO: 781 (P), 782 (P), 770 (P), 777 (P),
Sy. No.: SY NO: 781 (P), 782 (P), 770 (P), 777 (P),
(V): Chandampet
(M): Shankarampet (R)
(D): Medak
Pincode: 502255
Phone No: 9246169506

Lr.No.CEIG/TG/EI Ex. 650V/**HT-MDK-2025-10011**/D.No. 333/**2025** Dated: 31/01/2025

Sir,

Sub : The Electricity Act-2003 and Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023 - Electrical Installation of Voltage Exceeding 650 V of M/s **SRI RAJ IRON & STEEL PRODUCTS PVT LTD, Chandampet Shankarampet (R)** Mandal **Medak** District - Approval of Drawings, submitted towards Design Approval- Under Section 54 of The Electricity Act-2003 and Regulation - **45(3)** of CEA (Measures relating to Safety and Electric Supply) Regulations, 2023 Accorded Regarding.

Ref:- 1) Management letter received dated **29/01/2025**

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The drawings submitted for design approval are provisionally approved and valid up to **29/01/2026** subject to the compliance of conditions specified in the annexure to the drawings. The drawings approved for the erection of Equipment as per standard.

The defects/omission's pointed in the drawings will be specifically inspected by the Inspecting Officer at the time of Inspection. The Inspection fee of your installation works out to Rs. **52200** /- **Fifty Two Thousand Two Hundred** .

After completing the installation to your satisfaction, you are requested to submit the reports for inspection under Regulation **45(3)** of Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023.

The consumer shall provide the evidence that the supplier also completed his works in full shape, at the time of request for inspection of the installation.

The installation shall confirm to the relevant provisions of Central Electricity Authority (Measures relating to Safety and Electric Supply) Regulations, 2023 and Bureau of Indian Standards specifications.

Yours faithfully

CHIEF ELECTRICAL INSPECTOR TO GOVERNMENT

Annexure to Lr.No.CEIG /TS/ EI Ex. 650 V/D.No. 333 / 2025 Dt 31-01-2025

Provisionally approved subject to the compliance of the following conditions:

I. Sub-station

1. All electrical equipment shall be installed above the Highest Flood Level and where such equipment is not possible to be installed above Highest Flood Level, it shall be ensured that there is no seepage or leakage or logging of water.
2. where a substation or a switching station with oil-filled apparatus, such as transformer, static condenser, switchgear or oil circuit breaker having more than 2000 liters of oil is installed, whether indoors or outdoors
   a) Separation wall or fire barrier walls of thickness and dimensions as specified in the relevant standards shall be provided between the apparatuses and between the apparatus and adjacent building if building wall adjacent to the apparatuses is not rated for four hours fire withstand rating
   b) provisions shall be made for suitable oil soak pit and where use of more than 9000 litre of oil in any one oil tank, receptacle or chamber is involved, provision shall be made for the draining away or removal of any oil which may leak or escape from the tank, receptacle or chamber containing the same, and special precautions shall be taken to prevent the spread of any fire resulting from the ignition of the oil from any cause and adequate provision shall be made for extinguishing any fire which may occur.
3. Isolators/disconnectors and the controlling circuit breakers shall be inter-locked so that the isolators/disconnectors cannot be operated unless the corresponding breaker is in open position. (Regulation-47)
4. Isolators/disconnectors and the corresponding earthing switches shall be inter-locked so that no earthing switch can be closed unless and until the corresponding isolator/disconnector is in open position. (Regulation-47)
5. Maintain minimum safety working clearances specified in Schedule V of CEA Regulations for the bare conductors or live parts of any apparatus in outdoor substations excluding overhead lines of installations of voltage exceeding 650 V.
6. The entire switchyard or substation equipment and buildings including all non-current carrying metal parts associated with the installation shall be effectively earthed to an earthing system or mat. (Regulation-50)
7. Overhead lines, substations or generating stations shall be provided with surge arresters as per relevant standards for diverting electrical surges to the earth. The earthing lead for any lightning arrester shall be as short as possible and shall not pass through any iron or steel pipe, but shall be taken as directly as possible from the lightning arrester without touching any metal part to a separate vertical earth electrode or junction of the earth mat already provided subject to the avoidance of bends wherever practicable:

II. Transformers & Generators

1. Minimum safety clearances shall be maintained for all transformers and generators as per relevant standards specification so that sufficient space is available for easy operation and maintenance and for ensuring adequate ventilation (Regulation-46)

2. where two or more supplies are not intended to be operated in parallel, the respective circuit breakers or linked switches controlling the supplies shall be inter-locked to prevent possibility of any inadvertent paralleling or back feed. (Regulation-47)
3. All transformers and generators shall be installed as per relevant standards.
4. where two or more generators operate in parallel and neutral switching is adopted, inter-lock shall be provided to ensure that the generator breaker cannot be closed unless one of the neutrals is connected to the earthing system.
5. The neutral point of every generator and transformer shall be earthed with adequate size earthing conductor as per IS code 3043 by connecting it to the earthing system not by less than two separate and distinct connections.
6. Direct access to the transformer room shall be provided from outside and the surrounding walls of four hours fire withstand rating be provided as per relevant standards and the entrances to the transformer room shall be provided with fire resistant doors of two-hour fire rating and the door shall always be kept closed and a notice of this effect be affixed on outer side of the door
7. A curb of a suitable height be provided at the entrance in order to prevent the flow of oil from a ruptured transformer into other parts of the basement
8. The cables to primary side and secondary side have sealing at all floors and wall opening of atleast two hours fire withstand rating.
9. Only dry type transformer shall be used inside the residential and commercial buildings

III. Panels & Machinery Section

1. A clear space of not less than one meter in width shall be provided in front of the switchboard and wherever panels facing each other a clear space of not less than 2 m shall be maintained.
2. Ensure that all circuit breakers, cables and busbars are suitable for the prospective fault level.
3. All circuit breakers shall be selectively coordinated and ensure that the smallest cable connected to the circuit breaker is protected.
4. Earth fault or earth leakage protection shall be provided to disconnect the supply automatically, if the earth fault current exceeds the limit of current for keeping the contact potential within the reasonable values.
5. Wherever panels are installed on upper/lower floors of the building/production block, as isolating device/emergency stop button shall be provided on the ground floor in an easily access position to complete isolate the electric supply to the building/production block.
6. All panels, Motors, metal enclosures, metal structures supporting or associated with the installation, shall be connected with an earthing system as per IS code 3043 and the provisions of regulation 43.
7. All exposed conductive parts and extraneous conductive parts shall be earthed as per IS code 3043. (Regulation 43)
8. All electrical works shall be executed and required minimum clearances shall be maintained for easy operation and maintenance and ventilation purpose as per relevant standards, National Electric code (Sp30) and National Building code (SP7).

9. Automatic fire detection system and clean agent automatic fire extinguishing system shall be provided for all the panels, equipment and above the cable trays.

IV. General points

1. The material and apparatus used shall conform to the relevant standards and all electrical works shall be carried out as per relevant Indian standard.
2. All door openings from substation, electrical rooms, etc, should open outwards
3. All earth electrodes shall be constructed, and earthing work shall be carried out as per IS code 3043.
4. A minimum clearance of 3 m shall be maintained between earth electrodes and 1.5 m from nearest obstruction (i.e. Foundation of wall/building). All the earth pits shall be marked for easy identification.
5. Suitable arrangement shall be made for immediate and automatic or manual discharge of every static condenser on disconnection of supply.
6. All cables up to 33 KV and cable trenches shall be installed as per IS code 1255 and IS 1646.
7. Cables of exceeding 33 kV shall be laid with a minimum underground depth of 1.2 meter.
8. Cable trenches of more than 1000 cm^2 cross-sectional area shall be divided by incombustible barrier walls at intervals not exceeding 45 m as per IS code 1255.
9. Emergency lighting shall be provided in escape routes/passageways and in every attended station as per IS code 732.
10. Lightning protection shall be provided for the electrical installation as per the IS/IEC code 62305.
11. A residual current device shall be provided to disconnect the supply having rated residual current and duration as per IS code 732.
12. Manufacturer test reports/field test reports shall be submitted for the transformers, circuit breakers, equipment, cables etc.,
13. All earth electrodes test reports shall be submitted.
14. Separate load list for AC and DC Motor loads, MV and LV loads, Welding Loads, Heating loads etc., with details such as make, maker's S. No, capacity and voltage rating shall be submitted.

Yours faithfully

31.01.2025

for **Chief Electrical Inspector**
To Govt.of Telangana